

Vidio: Delivering a seamless, personalized viewing experience with Google Cloud ML

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With Google Cloud, Vidio is running a collaboration with Google Cloud to deliver a secure workplace, scalable, cost-effective infrastructure to support its OTT video service to more than 60 million Indonesian residents.

About Vidio

Vidio is a subsidiary of Emtek Group and provides a complete streaming service in Indonesia. As well as being connected directly to more than 21 television streaming services, Vidio presents thousands of local and international content to video-on-demand services and pay-television networks. Vidio is also among the leading apps downloaded by Indonesian residents from the Google Play Store and the App Store.

Industries: Media & Entertainment

Google Cloud results

- Scales effortlessly for high-traffic events to deliver a smooth user experience
- Improves developer productivity with automated deployments
- Accelerates business growth with cost-effective cloud compute resources

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Location: Indonesia

(<https://cloud.google.com/dataflow/>), [Cloud](#)
[Google Cloud](#) (<https://cloud.google.com/>)
[Kubernetes Engine](#)

[Cloud Pub/Sub](#) (<https://cloud.google.com/pubsub/>)
[Google Workspace](#)
[Dataflow](#) (<https://cloud.google.com/dataflow/>)

[Cloud Storage](#) (<https://cloud.google.com/storage/>)

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Video streaming is becoming increasingly entrenched in the Indonesian market and quality content is key to success. According to The Trade Desk, the Indonesian audience watches almost three billion hours

(https://pages.thetradedesk.com/connectedtv_SEA.html)

of over-the-top (OTT) per month. [Vidio](#)

(<https://www.vidio.com/>), a subsidiary of Emtek Group (Emtek), delivers premium content such as live sports and the latest drama series to subscribers on the web and through mobile phones and smart televisions. As one of the biggest over-the-top (OTT) providers in Indonesia with more than 60 Million Monthly Active Users (MAU), Vidio helps advertisers reach consumers at scale for their products and services.

Running its media streaming platform on Google Cloud (<https://cloud.google.com/>), Vidio has the agility to compete with industry players with its Subscription Video on Demand (SVOD) and an Advertising Video on Demand (AVOD) business model. Vidio uses a combination of Pub/Sub (<https://cloud.google.com/pubsub/docs/overview>), Dataflow (<https://cloud.google.com/dataflow>), and other technologies for media transcoding to stream content to viewers across devices. In addition, Vidio uses Cloud Storage (<https://cloud.google.com/storage>) as the data lake for its Emtek Digital customer data platform to help brand marketers improve ad targeting.

"Having multiple content offerings gives us much broader audience segments. Thus we need more superior data-driven user segmentation, and content curation by segments and AI-based recommendations so we can surface the right content recommendation for each user. Google Cloud provides us with the tools to support this requirement," says Sutanto Hartono, Managing Director at Emtek.

"We scaled our transcoding instances as needed to support Indonesia's dynamic news cycle. Within a short period of time, we grew our transcoding capacity by more than 9,000% so we can serve more viewers."

*—Tommy Sullivan, Chief
Technology Officer, Vidio*

Improving user experience for high traffic events

Although Vidio offers free VOD content, paid premium content such as its live sports events drive revenue and support the business. However, the fluctuation in consumption patterns becomes a

challenge when delivering live streaming content. If the platform cannot handle traffic spikes, viewers may not be able to sign up or watch the content, impacting the overall user experience.

During the Olympics Games, the difference between low and peak traffic was 11 times higher, from thousands of requests per second to 500,000 when viewers signed up at the last minute to watch Indonesia compete in the badminton finals, for example. Like the Olympic Games, Vidio expects high demand from local football fans for the World Cup 2022 broadcast.

Leveraging Google Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine>) (GKE) for auto-scaling allows Vidio to handle more traffic faster and deliver a positive experience. Vidio is in the process of migrating its 33 microservices to GKE.

"Google Kubernetes Engine gives us more fine-grained control in terms of auto-scaling and is more efficient for CPU usage. Our goal is to achieve 50% CPU utilization, so we can optimize server use and invest in other value-added projects."

At the same time, GKE accelerates innovation by driving developer productivity. Using automated deployment, engineers can set up environments for new workloads in just one day.

"We found Google Cloud to be very cost efficient. So we moved our data, computing, and media files to the platform as part of a broader plan to standardize our usage."

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Increasing click-through with content recommendations

Vidio needs to engage Indonesian viewers with relevant content, including local drama and sports events, to provide a service that is on par with multinational OTT providers. This makes discoverability a key responsibility of the Vidio engineering team. Applying machine-learning algorithms on Google Cloud and MLflow open-source platform to analyze customer data, Vidio can

predict what they want to watch based on their preferences and behavior.

"We have millions of professionally produced content and it is extremely important we surface the right content for each user," says Tommy Sullivan, Chief Technology Officer at Vidio "We do this through a balance of data-driven user segmentation, per-segment content curation, and machine learning for behavioral recommendations. Delivering that content quickly, efficiently, and reliably is another set of engineering responsibilities."

With Click-Through Rates (CTR) on [Google Analytics](https://analytics.google.com/analytics/web/) (<https://analytics.google.com/analytics/web/>), the data science team can measure how many viewers click on the recommended content and fine-tune the algorithms to improve accuracy. Similarly, the team measures the effectiveness of push notifications of new content to users to ensure a positive user experience.

Driving cost-efficiency with preemptible cloud resources

Vidio started operations on a domestic cloud service in Indonesia but quickly found its demands exceeding the provider's capacity. "There were instances where our demands—video being high bandwidth and high throughput—maxed out the cloud service's capacity," says Sullivan. "We simply weren't able to buy any more capacity from them."

The business then opted to use the infrastructure provided by a traditional multinational cloud service.

However, upon benchmarking Google Cloud pricing and compute efficiency against other cloud providers, Sullivan says they found Google Cloud to be very cost efficient. "So we moved our data, computing, and media files to the platform as part of a broader plan to standardize our usage."

Controlling compute costs for transcoding

Vidio is also controlling its compute costs by using preemptible instances such as affordable, short-lived virtual machine instances optimized for batch jobs and fault-tolerant applications. It found Google Cloud provides better value than what other cloud providers offer. The business is also able to use those instances to complete transcoding (converting video to the best format for its service) in a scalable infrastructure.

Using Pub/Sub for event ingestion and delivery and Dataflow for fully managed stream analytics processing, Vidio has dramatically improved its transcoding capacity to support increased demand during high-profile events or incidents. For VOD, videos are uploaded to Cloud Storage, divided, and transcoded in five-minute segments into different bit-rates so viewers can stream from mobile phones and smart televisions. Using multiple preemptible instances on Google Cloud, Vidio can transcode faster and cost-effectively.

"We scaled our transcoding instances as needed to support Indonesia's dynamic news cycle," says Sullivan. "Within a short period of time, we grew our

transcoding capacity by more than 9,000% so we can serve more viewers."

Transforming meeting productivity with Google Workspace

Prior to becoming a Google Cloud customer, employees at Vidio were already using Google Workspace (https://workspace.google.com/intl/en_sg/).

Now, it is able to value add by connecting Google Cloud access control to the Google Workspace platform. "Many companies use Google Workspace's Groups for Business for communication and workspace collaboration, but having those same access control lists flow into Google Cloud services has made my life easier," says Sullivan.

"Our employees can move seamlessly between departments and Groups, while identity and endpoint administration available through Google Workspace enables us to manage all of them from a single console."

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Improving ad effectiveness with personalized marketing

Vidio launched its Emtek Digital customer data platform on Google Cloud to serve its AVOD business. Hosted in the Indonesia cloud region for data sovereignty, the CDP analyzes behavioral data from more than 100 million monthly users who interact with Emtek entities ecommerce marketplace Bukalapak, digital media company KapanLagi, and Vidio. Emtek Digital helps the sales team query the data and build customer segments based on what the users watch and share so advertisers can promote their products or services to the desired audience.

As the business grows, Cloud Storage provides Vidio with the scalability to ingest and process more data over time. The data lake currently stores 660 TB of user data such as play, pause, and recommendations, and streams more than 2 billion events a day through Pub/Sub.

"Google Cloud accelerates our transformation as a data company," says Sullivan. "We not only record data such as error rates but more importantly, we're pulling insights from data to delight viewers with great experiences."